



Cambridge IGCSE™

CANDIDATE
NAMECENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

MATHEMATICS

0580/22

Paper 2 Non-calculator (Extended)

October/November 2025

2 hours

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages.

List of formulas

Area, A , of triangle, base b , height h .

$$A = \frac{1}{2}bh$$

Area, A , of circle of radius r .

$$A = \pi r^2$$

Circumference, C , of circle of radius r .

$$C = 2\pi r$$

Curved surface area, A , of cylinder of radius r , height h .

$$A = 2\pi rh$$

Curved surface area, A , of cone of radius r , sloping edge l .

$$A = \pi rl$$

Surface area, A , of sphere of radius r .

$$A = 4\pi r^2$$

Volume, V , of prism, cross-sectional area A , length l .

$$V = Al$$

Volume, V , of pyramid, base area A , height h .

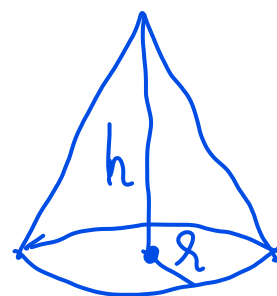
$$V = \frac{1}{3}Ah$$

Volume, V , of cylinder of radius r , height h .

$$V = \pi r^2 h$$

Volume, V , of cone of radius r , height h .

$$V = \frac{1}{3}\pi r^2 h$$



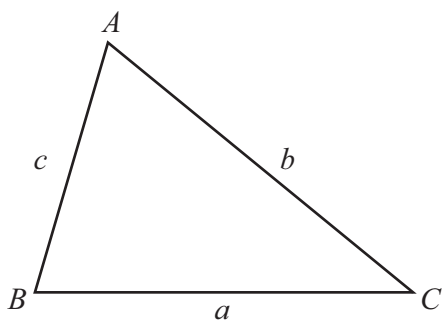
Volume, V , of sphere of radius r .

$$V = \frac{4}{3}\pi r^3$$

For the equation $ax^2 + bx + c = 0$, where $a \neq 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For the triangle shown,



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area} = \frac{1}{2}ab \sin C$$



Calculators must **not** be used in this paper.

- 1 The temperature at 4 am is -12°C .
The temperature at 4 pm is 21°C .

Find the increase in temperature from 4 am to 4 pm.

..... $^{\circ}\text{C}$ [1]

- 2 Find **all** the common factors of 8 and 12.

..... [1]

- 3 A cuboid has length 6 cm, width 5 cm and height 2.5 cm.

Work out the volume of the cuboid.

..... cm^3 [2]

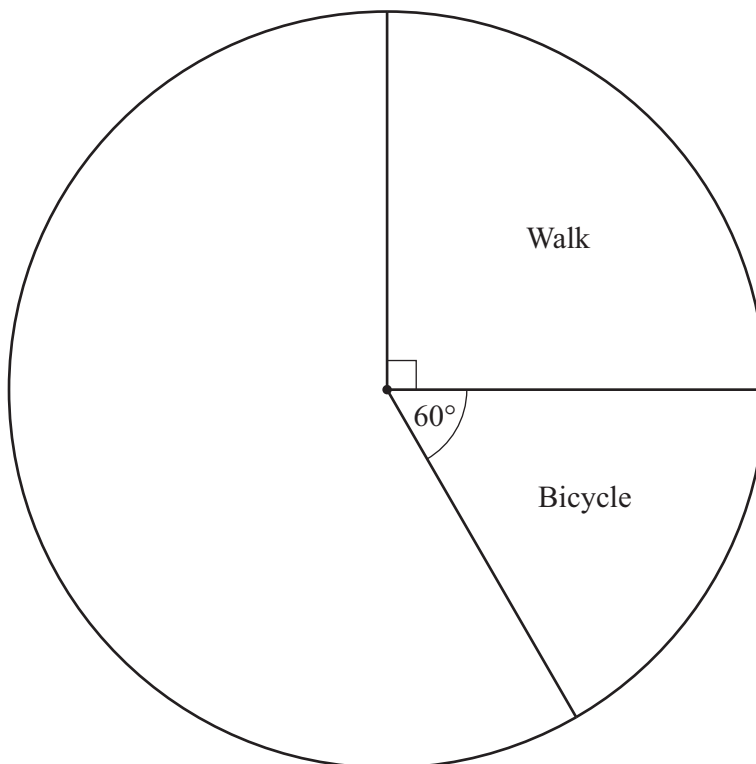
- 4 A computer costs \$560.
In a sale, this cost is reduced by 20%.

Find the cost of the computer in the sale.

\$ [2]



- 5 The pie chart shows some information about the way 600 students travel to school.



- (a) Work out the number of students that walk to school.

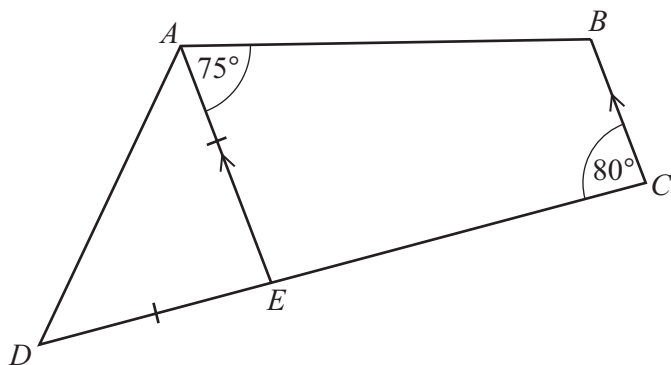
..... [2]

- (b) 120 of the students travel to school by car.
The remaining students travel by bus.

Complete the pie chart.

[3]





NOT TO
SCALE

$ABCD$ is a quadrilateral.
 E lies on CD and AE is parallel to BC .
 $EA = ED$.

Find

(a) angle ABC

Angle $ABC = \dots\dots\dots$ [1]

(b) angle AED

Angle $AED = \dots\dots\dots$ [1]

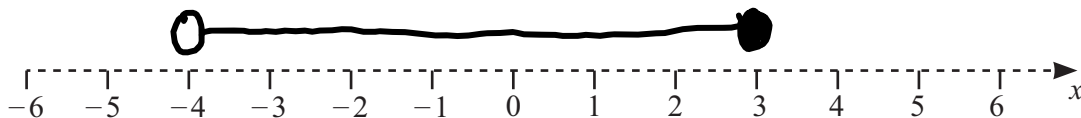
(c) angle DAB .

Angle $DAB = \dots\dots\dots$ [2]

biết những gì

7 Represent the inequality $-4 < x \leq 3$ on the number line.

(có dấu =)



[2]



$$1\$ = 100 \text{ cents}$$

- 8 Kemi buys p pens that each cost 40 cents.

She pays with \$20. = 2000 cents

Write an expression, in terms of p , for the change, in cents, Kemi receives from the \$20.

Money of p pens = $40p$ cents
 $\rightarrow$ Kemi receive = $2000 - 40p$
 $\textcircled{1} = 100\%$ $2000 - 40p$ cents [2]

- 9 Rajid has a full bottle of juice.

He drinks $\frac{1}{3}$ of the full bottle on Monday.

He drinks $\frac{3}{7}$ of the full bottle on Tuesday.

Find the fraction of the bottle of juice remaining.

Monday and Tuesday he drinks: $\frac{1}{3} + \frac{3}{7} = \frac{16}{21}$
 $\rightarrow$ Remaining = $1 - \frac{16}{21} = \frac{5}{21}$

10

$$b = dm + 2mk$$

$$\rightarrow b = m(d + 2k) \rightarrow m = \frac{b}{d + 2k}$$

- (a) $d = 3.14$, $m = 7.92$ and $k = 10.16$.

By rounding each value correct to 1 significant figure, work out an estimate for b .

$$d = 3 \quad m = 8 \quad k = 10$$

$$b = 3 \times 8 + 2 \times 8 \times 10 = 24 + 160 = 184$$

- (b) Rearrange the formula to make m the subject.

$$V(\text{cm}^3)$$

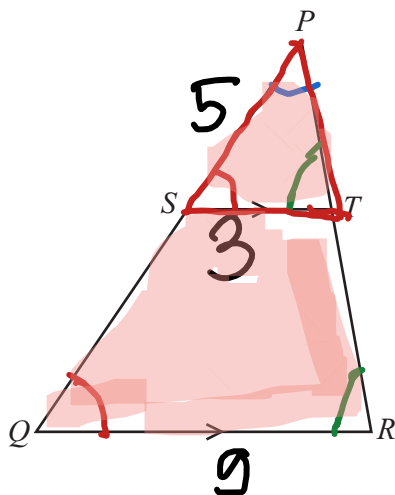
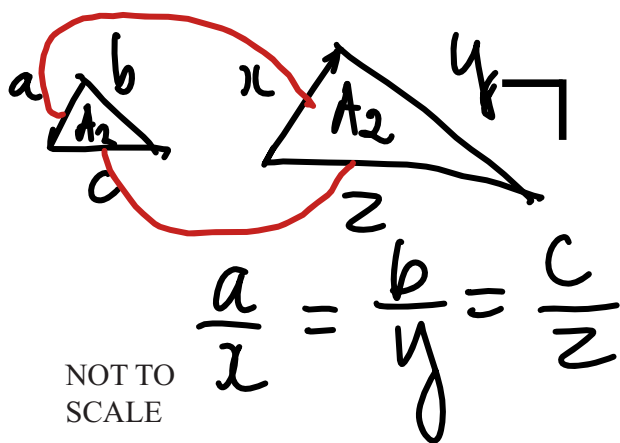
$$A(\text{cm}^2)$$

$$\frac{V_1}{V_2} = \left(\frac{a}{x}\right)^3$$

$$m = \frac{b}{d + 2k}$$

$$\frac{A_1}{A_2} = \left(\frac{a}{x}\right)^2$$





In the diagram, S lies on PQ and T lies on PR .
 ST is parallel to QR .

- (a) Explain why triangle PQR is mathematically similar to triangle PST .
 Give a geometrical reason for each statement you make.

angle P common angle
 angle $PST =$ angle PQR
 (corresponding angle)
 angle $PTS =$ angle PRQ [3]

- (b) $ST = 3$ cm, $QR = 9$ cm and $PS = 5$ cm.

Work out PQ .

$$\frac{3}{9} = \frac{5}{PQ} \rightarrow PQ = 15$$

$PQ = 15$ cm [2]

- (c) The area of triangle PST is $2k$ cm².

Find, in terms of k , the area of quadrilateral $QRTS$.

$$\frac{\text{Area } \triangle PST}{\text{Area } \triangle PQR} = \left(\frac{3}{9}\right)^2$$

$$\rightarrow \frac{2k}{\text{Area } \triangle PQR} = \frac{1}{9} \rightarrow \text{Area } \triangle PQR = 18k$$

- 12 A fitness club has 100 members.

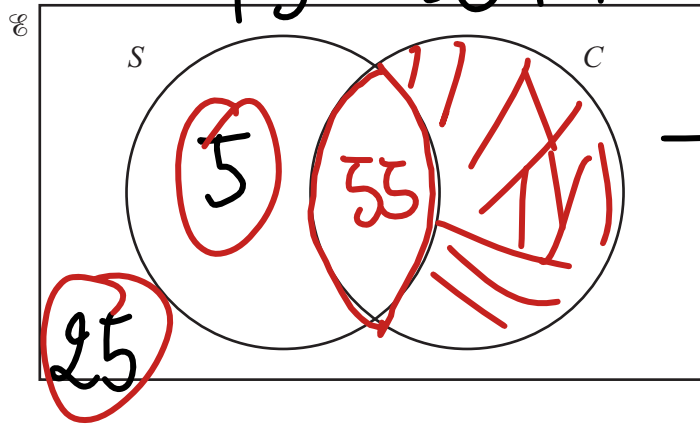
60 swim (S).

70 cycle (C).

25 do not swim or cycle.

$$75 = S + C - S \text{ and } C$$

$$75 = 60 + 70 - S \text{ and } C$$



$$\rightarrow S \text{ and } C = 55$$

- (a) Complete the Venn diagram.

[3]

- (b) One member of the fitness club is chosen at random.

For this member, find

(i) $P(C) = \frac{70}{100}$

..... [1]

(ii) $P(S \cap C) = \frac{55}{100}$

..... [1]

$C' = \text{not } C$

(iii) $P(S \cup C') = \frac{85}{100}$

..... [1]

cube cube cube

$(b^2)^3$

$(c^3)^3$

$a^3 \times b^6 \times c^9$ cube number

13

$$M = 2^7 \times 3^3 \times 5^2$$

$$\rightarrow \frac{M}{R} =$$

$$\frac{2^7 \times \boxed{3^3} \times 5^2}{\boxed{2^1 \times 5^2} \rightarrow R = 50}$$

- (a) Write $14M$ as a product of its prime factors.
Give your answer in index form.

$$14M = 7 \times 2 \times 2^7 \times 3^3 \times 5^2$$

$$= \underline{7 \times 2^8 \times 3^3 \times 5^2} \quad (7, 2, 3, 5: \text{Prime number})$$

..... [2]

- (b) R is an integer.

$\frac{M}{R}$ is a cube number.

Indices

Find the smallest possible value of R .

$$\left| \begin{array}{l} \frac{a^m}{a^n} = a^{m-n} \\ (1) \end{array} \right. \quad a^{-m} = \frac{1}{a^m} \quad (2) \quad \sqrt[m]{a^n} = a^{\frac{n}{m}} \quad (3)$$

$$R = \dots 9 \times 9 \dots [2]$$

- 14 Find the value of

- (a) $3^2 \div 3^{-2}$

$$= \frac{3^2}{3^{-2}} = 3^4 = \overbrace{3 \times 3 \times 3 \times 3}^{9 \times 9} = 81$$

- (b) $16^{-\frac{3}{2}}$

$$= \frac{1}{16^{\frac{3}{2}}} = \frac{1}{\sqrt[2]{16^3}} = \frac{1}{(\sqrt[2]{16})^3} \quad [2]$$

$$= \frac{1}{4^3} = \frac{1}{64} \quad [2]$$



10

$$a^2 - b^2 = (a-b)(a+b)$$

15 Factorise.

(a) $x^2 - 64$

$$= x^2 - 8^2 \\ = (x-8)(x+8)$$

(b) $5x(x-2y) + 6(x-2y)^2$

$$= (x-2y)(5x + 6(x-2y)) = (x-2y)(5x + 6x - 12y) = (x-2y)(11x - 12y)$$

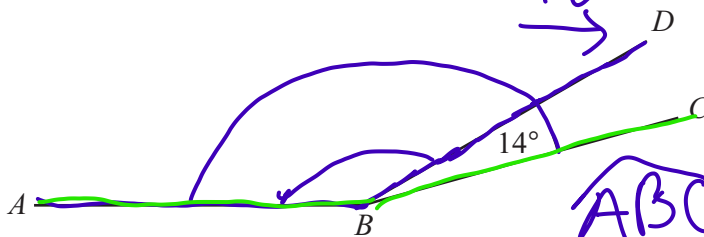
[1]

Set up equation contain n $\widehat{ABC} - \widehat{ABD} = 14$

$$\frac{180(n-2)}{n} - 156 = 14$$

[2]

$$n = 36$$



NOT TO SCALE

AB and BD are two sides of a regular 15-sided polygon.

AB and BC are two sides of a regular n -sided polygon.

Angle $DBC = 14^\circ$.

Work out the value of n .

$$\widehat{ABC} = \frac{180 \times (n-2)}{n}$$

$$\widehat{ABD} = \frac{180 \times (15-2)}{15} = 156$$

$$m_1 \text{ (gradient)} \quad m_2 \text{ (gradient)}$$

$$m_1 \times m_2 = -1 \rightarrow m_2 = -\frac{1}{m_1}$$

$$m_1 = m_2$$

$$m = \frac{1}{4}$$

pass through point $M(-4, 5)$

$$\text{Gradient } BD = \frac{y_D - y_B}{x_D - x_B} = \frac{5 - 1}{-3 - (-3)} = \frac{4}{-2} = -2$$

[4]

$$\rightarrow \text{gradient } AC : m = \frac{1}{-2} = -\frac{1}{2}$$

$$y = \frac{1}{4}x + c \quad (y = mx + c)$$

$$\text{sub } M(-4; 5) \rightarrow 5 = \frac{1}{4} \times (-4) + c$$

- 17 B is the point $(-3, 1)$ and D is the point $(-5, 9)$.
BD is a diagonal of the kite ABCD.

- (a) The ratio of the lengths of the diagonals $BD : AC = 2 : 3$.

Work out the length of AC.

Give your answer as a surd in its simplest form.

$$y = \frac{1}{4}x + 6$$

$$\rightarrow c = 6$$

$$BD = \sqrt{(-3+5)^2 + (1-9)^2}$$

$$BD = \sqrt{2^2 + (-8)^2}$$

$$= \sqrt{4 + 64} = \sqrt{68}$$

$$\frac{\sqrt{68}}{AC} = \frac{2}{3}$$

$$\rightarrow 2AC = 3\sqrt{68}$$

$$\rightarrow AC = \frac{3\sqrt{68}}{2}$$

[5]

- (b) Find the coordinates of the midpoint of BD.

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\left(\frac{-3 + -5}{2}, \frac{1 + 9}{2} \right) \quad [2]$$

- (c) The diagonal AC of the kite passes through the midpoint of BD.

Find an equation of AC.

Give your answer in the form $y = mx + c$.

$$= (-4; 5)$$

$$y = \dots\dots\dots [4]$$

12

18 Rationalise the denominator and simplify.

(Remove square root of denominator)

$$(a+b)(a-b) = a^2 - b^2$$

$$\frac{20}{4 + \sqrt{6}}$$

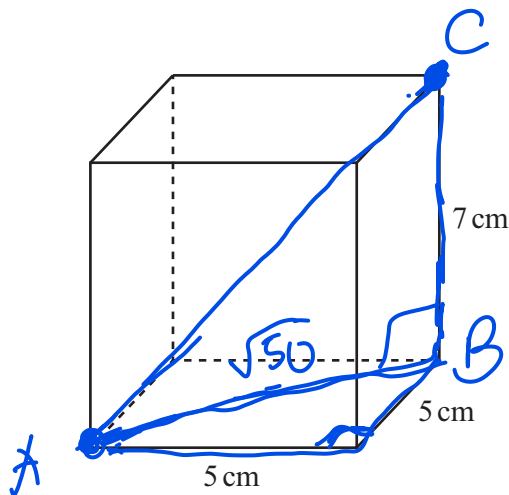
$$\begin{aligned} &= \frac{20 \times (4 - \sqrt{6})}{(4 + \sqrt{6}) \times (4 - \sqrt{6})} \\ &= \frac{20 \times (4 - \sqrt{6})}{4^2 - (\sqrt{6})^2} \\ &= \frac{20(4 - \sqrt{6})}{16 - 6} \\ &= \frac{20(4 - \sqrt{6})}{10} \end{aligned}$$

[3]

$$(\sqrt{a})^2 = a$$

NOT TO SCALE

19



The diagram shows a box in the shape of a cuboid.
Mala has a straight rod of length 10 cm.

Show that this rod does **not** fit completely inside the box.

$$AB = \sqrt{5^2 + 5^2} = \sqrt{25 + 25} = \sqrt{50}$$

$$\triangle ABC: AC = \sqrt{(\sqrt{50})^2 + 7^2} = \sqrt{50 + 49}$$

$$\text{Length of rod: } 10 > AC = \sqrt{99}$$

[3]

$$\rightarrow \text{Not fit (bs rod} > AC)$$

20

$$f(x) = \frac{21}{2x-1}, x \neq \frac{1}{2}$$

$$g(x) = 3x + 4$$

(a) Find

(i) $g(2) = 10$

..... [1]

(ii) $gf(-1) =$

$$f(-1) = \frac{21}{-3} = -7 \quad g(-7) = -17$$

..... [2]

(iii) $f^{-1}(x).$

$$y = \frac{21}{2x-1} \rightarrow 2x-1 = \frac{21}{y} \rightarrow 2x = \frac{21}{y} + 1$$

$$y \rightarrow x = \frac{21}{2y} + \frac{1}{2}$$

$f^{-1}(x) = \frac{21}{2x} + \frac{1}{2}$ [3]

(b) Solve $f(x) = g(x).$

$$\frac{21}{2x-1} = 3x+4$$

$$f^{-1}(x) = \frac{21}{2x} + \frac{1}{2}$$

$$21 = (3x+4)(2x-1)$$

$$21 = 6x^2 - 3x + 8x - 4$$

$$0 = 6x^2 + 5x - 25$$

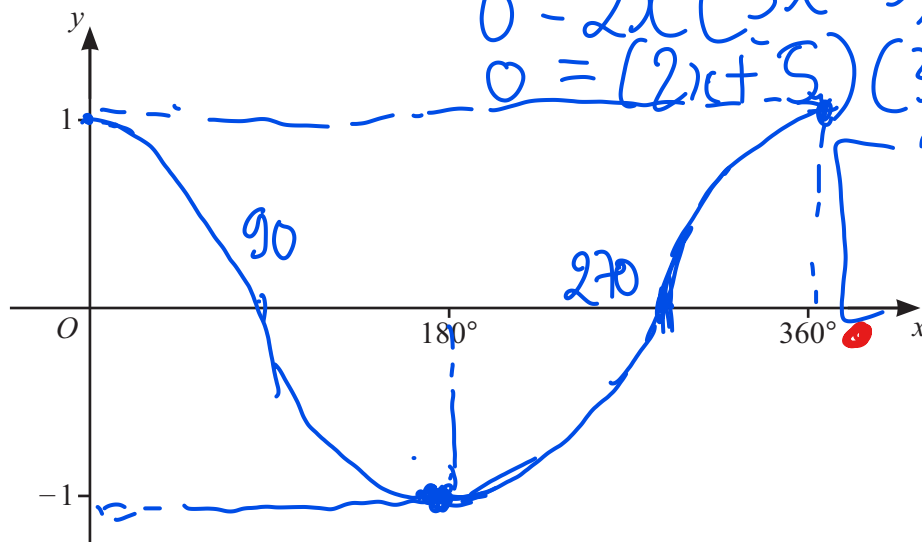
Factoring: $xy = -150$ (a x c) -5
 $x+y = 5$ (b)
 $-10, 15$

$$x = \frac{-5}{2} \text{ or } x = \frac{5}{3}$$

[5]



21 (a)



14

$$\begin{aligned}
 0 &= 6x^2 - 10x + 15x - 25 \\
 0 &= 2x(3x - 5) + 5(3x - 5) \\
 0 &= (2x + 5)(3x - 5) \\
 2x + 5 &= 0 \\
 3x - 5 &= 0
 \end{aligned}$$

On the diagram, sketch the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$.

[2]

- (b) Solve the equation $2\cos x + \sqrt{3} = 0$ for $0^\circ \leq x \leq 360^\circ$

$$y = ax^2 + bx + c$$

turning point (max, min) = $\left(\frac{-b}{2a}; \text{sub vào } y \right)$

$x = \frac{150}{2} \text{ or } x = \frac{210}{2}$ [3]

- 22 A graph with equation $y = x^2 + bx + c$ has a minimum point at $(-5, 12)$.

Find the value of b and the value of c .

$$-5 = \frac{-b}{2a}$$

$$b = 10$$

$$-5 = \frac{-b}{2 \times 1} \rightarrow -10 = -b$$

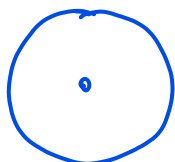
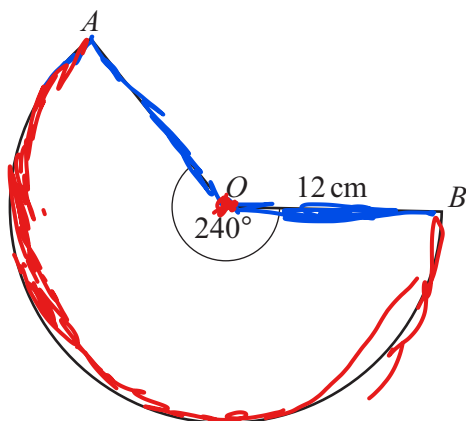
$$b = \frac{10}{37} + 3$$

$$c = \frac{37}{37} \text{ [3]}$$

$$\begin{aligned}
 12 &= (-5)^2 + 10x - 5 + c \\
 12 &= 25 - 50 + c = 37 \\
 &= -25 + c \rightarrow c = 12 + 25
 \end{aligned}$$



23

 πr^2 $2\pi r$ 

The diagram shows a major sector, AOB , of a circle.
The sector angle is 240° and the radius is 12 cm .

- (a) Show that the length of the major arc AB is $16\pi\text{ cm}$.

$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r = \frac{240}{360} \times 2\pi \times 12$$

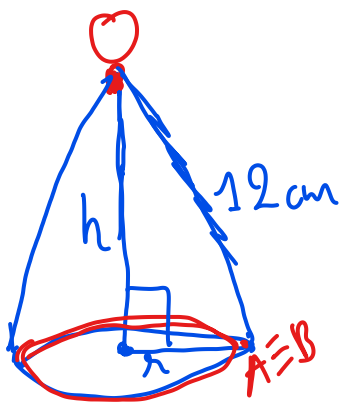
$$= 16\pi$$

[1]

- (b) OA is joined to OB to form a cone.

Work out the volume of the cone.

Give your answer in the form $\frac{(a\sqrt{b})\pi}{3}$ where a is an integer and b is a prime number.



$$V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \times 64 \times 4\sqrt{5}$$

$$2\pi r = 16\pi \rightarrow r = \frac{16\pi}{2\pi} = 8$$

$$12^2 = h^2 + 8^2$$

$$144 = h^2 + 64 \rightarrow h^2 = 144 - 64$$

$$h^2 = 80$$

$$\rightarrow h = \sqrt{80} = 4\sqrt{5}$$

cm³ [6]

Question 24 is printed on the next page.

24

$$27^{nx} = (9^x)^2$$

Find the value of n .

Indices

$$= (3^{2x})^2$$

$$(3^3)^{nx} = 3^{4x}$$

$$3^{3nx} = 3^{4x}$$

$$\rightarrow 3nx = 4x$$

$$\rightarrow 3n = 4 \rightarrow n = \frac{4}{3}$$

$$n = \frac{4}{3} \quad [2]$$

$$3^x = 3^y$$

$$\rightarrow x = y$$

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

